Unit	Question *
1	Explain the concept of project identification in the project life cycle.
1	Describe the purpose of market and demand analysis in a feasibility study.
1	Explain the difference between technical feasibility and financial feasibility.
1	Summarize the objectives of conducting a project feasibility study.
1	Define project cost estimation.
1	List the different methods of financial appraisal.
1	Explain the concept of payback period.
1	Describe the purpose of financial appraisal in project selection.
1	Define PERT and CPM.
1	List the differences between PERT and CPM.
1	Explain the concept of an activity network diagram.
1	Describe the terms 'event' and 'activity' in network analysis.
1	Define critical path.
1	Explain the concept of precedence relationship in network diagrams.
1	List the rules for numbering events in a network diagram.
1	Describe forward pass and backward pass calculations.
1	Define total float and free float.
1	Explain the concept of crashing in project scheduling.
1	List the types of float used in network analysis.
1	Describe the relationship between cost and time in project crashing.
2	Define project cost control.
2	Explain the concept of the PERT/Cost system.
2	Describe the purpose of Budgeted Cost of Work Scheduled (BCWS).
2	List the components considered in project cost control.
2	Define resource scheduling.
2	Explain the concept of resource leveling.
2	List the objectives of resource allocation in projects.
2	Describe resource smoothing.
2	Define project risk.
2	List the types of risks encountered in project management.
2	Explain the concept of risk identification.
2	Describe the purpose of a risk register.
2	Define project control.
2	List the various project control techniques.
2	Explain the concept of earned value analysis.
2	Describe the purpose of milestone tracking.
2	Define project audit.
2	Explain the concept of project termination.
2	List the types of project termination.
2	Describe the purpose of a post-project review.
3	Define Agile project management.
3	List the key characteristics of Agile methodology.
3	Explain the difference between traditional and Agile project management.
3	Describe the Agile Manifesto.

3	List the 12 principles of Agile.
3	Explain the principle of customer collaboration in Agile.
3	Describe the concept of self-organizing teams.
3	Define the principle of sustainable development in Agile.
3	List different Agile methodologies.
3	Define Extreme Programming (XP).
3	Explain the concept of Kanban.
3	Describe Feature Driven Development (FDD).
3	Define Lean software development.
3	Explain the relationship between Agile and Scrum.
3	List the similarities between Lean and Agile.
3	Describe how DevOps complements Agile practices.
3	Define ITIL.
3	List the objectives of IT service management.
3	Explain the relationship between Agile and ITIL.
3	Describe the concept of service lifecycle in ITIL.
4	Define Sprint in Scrum.
4	Explain the concept of Product Backlog.
4	List the key Scrum terminologies.
4	Describe the purpose of a Sprint Backlog.
4	Define Sprint Review.
4	Explain the purpose of a Sprint Backlog.
4	List the participants of a Sprint Review meeting.
4	Describe the outcome of a Sprint Review.
4	Define Sprint Retrospective.
4	Explain the purpose of a retrospective meeting.
4	List the common formats used for conducting a retrospective.
4	Describe the outcome of a Sprint Retrospective.
4	List the three roles in Scrum.
4	Define the role of Scrum Master.
4	Explain the responsibilities of a Product Owner.
4	Describe the role of the Development Team in Scrum.
4	List any four best practices of Scrum.
4	Define Definition of Done in Scrum.
4	Explain the concept of a Scrum board.
4	Describe the significance of the Daily Scrum meeting.
5	Define DevOps.
5	List the key components of DevOps.
5	Explain the relationship between Development and Operations in DevOps.
5	Describe the goals of DevOps culture.
5	Define containerization.
5	Explain the concept of a Docker image.
5	List the advantages of using Docker containers.
5	Describe the difference between a container and a virtual machine.
5	Define Continuous Integration (CI).

5	Explain the concept of version control.
5	List the benefits of automated builds.
5	Describe the purpose of a CI pipeline.
5	Define Test Driven Development (TDD).
5	Explain the concept of automated testing.
5	List the types of automated tests used in software development.
5	Describe the Red-Green-Refactor cycle in TDD.
5	Define configuration management.
5	Explain the purpose of configuration management tools.
5	List examples of configuration management tools.
5	Describe the concept of Infrastructure as Code (IaC).
5	Define Continuous Deployment.
5	Explain the difference between Continuous Delivery and Continuous Deployment.
5	List the benefits of automated monitoring.
5	Describe the purpose of a monitoring dashboard.

Unit	Question *
1	Analyze the various types of feasibility studies conducted before approving a project.
1	Evaluate the importance of market and demand analysis in reducing project risk.
1	Analyze how technical feasibility influences the selection of project technology.
1	[Scenario-Based] A company is planning to launch a new product. Analyze the feasibility factors it must evaluate before proceeding.
1	Evaluate the consequences of skipping feasibility analysis on project success.
1	Analyze the roles of primary and secondary data in market/demand analysis.
1	Analyze the various techniques used for estimating project costs.
1	Evaluate the significance of Net Present Value (NPV) in financial appraisal.
1	Analyze the limitations of the payback period method in investment decisions.
1	[Scenario-Based] A firm has two competing project proposals. Evaluate how financial appraisal techniques would help select the better one.
1	Analyze the relationship between cost estimation accuracy and project success.
1	Evaluate the impact of inflation and time value of money on financial appraisal.
1	Analyze the differences between PERT and CPM with respect to their applications.
1	Evaluate the suitability of PERT for projects with uncertain activity durations.
1	Analyze the steps involved in constructing a project network diagram.
1	[Scenario-Based] A construction project has multiple interdependent activities. Analyze how a network diagram would help schedule it.
1	Evaluate the advantages and limitations of using CPM for large-scale projects.
1	[Scenario-Based] A hospital is setting up a new diagnostic wing with several parallel and sequential tasks. Analyze how the three-time PERT estimates would help plan this project.
1	Analyze the procedure for determining the critical path in a project network.
1	Evaluate the importance of precedence relationships in accurate scheduling.
1	Analyze how forward and backward pass calculations determine earliest and latest event times.
1	[Scenario-Based] Given a project with a list of activities, durations and dependencies, analyze the process to identify its critical path.
1	Evaluate the impact of an incorrect precedence relationship on the overall project schedule.
1	Analyze the significance of the critical path in project time management.
1	Analyze the procedure for calculating total float, free float and independent float.
1	Evaluate the cost-time trade-off involved in crashing project activities.
1	Analyze the criteria for selecting activities to crash in a project network.
1	[Scenario-Based] A project is running behind schedule. Analyze how crashing selected activities could help meet the deadline at minimum additional cost.
1	Evaluate the risks associated with excessive crashing of project activities.
1	Analyze how float values help project managers prioritize non-critical activities.
2	Explain the working of the PERT/Cost system in monitoring project expenditure.
2	Apply the concept of cost control to identify budget deviations in a project.
2	Illustrate the process of integrating cost and time in project control.
2	[Scenario-Based] A project has exceeded its budgeted cost at the halfway stage. Apply appropriate cost control techniques to bring it back on track.
2	Explain the various cost variance measures used in project cost control.
2	Apply PERT/Cost analysis to evaluate the financial performance of a sample project.
2	Explain the steps involved in resource scheduling for a project.
2	Apply resource leveling techniques to resolve resource over-allocation in a project.
2	Illustrate the difference between resource leveling and resource smoothing with examples.
2	[Scenario-Based] A project manager finds two critical activities require the same resource at the same time. Apply resource leveling to resolve the conflict.
2	Explain the impact of limited resources on project duration.
2	Apply resource allocation techniques to optimize the use of manpower in a project.
2	Explain the steps involved in the project risk management process.
2	Apply qualitative risk analysis techniques to prioritize risks in a project.
2	Illustrate the difference between qualitative and quantitative risk analysis.
2	[Scenario-Based] A software project faces the risk of key personnel leaving mid-project. Apply risk mitigation strategies to handle this situation.
2	Explain the various risk response strategies used in project management.
2	Apply risk monitoring techniques to track identified risks throughout a project.
2	Explain the significance of earned value management in project control.
2	Apply earned value analysis to determine the schedule and cost performance of a project.
2	Illustrate the use of Gantt charts as a project control tool.
2	[Scenario-Based] A project shows a negative cost variance at the review stage. Apply appropriate control techniques to correct the deviation.
2	Explain the role of periodic status reports in project control.
2	[Scenario-Based] Apply control charts to monitor the performance of ongoing project activities.
2	Explain the objectives and scope of a project audit.
2	Apply the project audit process to evaluate the performance of a completed project.
2	Illustrate the different circumstances under which a project may be terminated.
2	[Scenario-Based] A project is significantly behind schedule and over budget with little chance of success. Apply appropriate decision-making steps to determine whether to terminate it.
2	Explain the steps involved in the project closure process.
2	Apply lessons-learned documentation techniques to improve future project performance.
3	Explain the core values outlined in the Agile Manifesto.
3	Apply Agile principles to justify the adoption of Agile in a software development project.
3	Illustrate the differences between the Waterfall model and Agile project management.
3	[Scenario-Based] A company transitioning from Waterfall to Agile struggles with adapting its planning process. Apply Agile concepts to address this issue.
3	[Scenario-Based] Explain the benefits and challenges of adopting Agile project management.
3	Apply the concept of iterative development to explain how Agile handles changing requirements.
3	Explain the significance of customer collaboration as an Agile principle.
3	[Scenario-Based] Apply the principle of welcoming changing requirements to a real-world software project scenario.
3	Illustrate how self-organizing teams contribute to Agile project success.
3	Apply the principle of continuous delivery to explain its benefits to project stakeholders.
3	Explain the importance of face-to-face communication as emphasized in Agile principles.
3	Apply Agile principles to evaluate the working practices of a hypothetical development team.
3	Explain the key practices followed in Extreme Programming.
3	[Scenario-Based] Apply Kanban principles to manage the workflow of an IT support team.
3	Illustrate the differences between Scrum and Kanban methodologies.
3	Apply the concept of Feature Driven Development to plan a small software module.
3	Explain the role of pair programming in Extreme Programming.
3	Apply Lean software development principles to reduce waste in a project.
3	Explain how Scrum implements the principles of Agile methodology.
3	Apply the concepts of Lean and Agile to identify wasteful practices in a project.
3	Illustrate the relationship between Agile development and DevOps deployment practices.
3	[Scenario-Based] An organization practices Agile but faces delays in deployment. Apply DevOps concepts to bridge this gap.
3	Explain the differences between Lean manufacturing principles and Lean software development.
3	Apply the combined Agile-DevOps approach to improve software delivery speed.
3	Explain the key stages of the ITIL service lifecycle.
3	Apply Agile concepts to improve the incident management process defined in ITIL.
3	Illustrate how Agile and ITIL can be integrated in an IT organization.
3	[Scenario-Based] An IT support team using ITIL processes wants to adopt Agile sprints for service improvement. Apply Agile-ITIL integration to design this approach.

3	Explain the challenges of combining Agile practices with traditional ITIL processes.
3	Apply change management principles from ITIL to an Agile project environment.
4	Explain the process of creating and refining a Product Backlog.
4	Apply Scrum terminology to describe how a Sprint is planned and executed.
4	Illustrate the relationship between the Product Backlog and the Sprint Backlog.
4	[Scenario-Based] A Scrum team receives new customer requirements mid-sprint. Apply Scrum principles to decide how this should be handled.
4	Explain the concept of Sprint Goal and its importance.
4	Apply backlog prioritization techniques to organize a sample product backlog.
4	Explain the process followed during a Sprint Review meeting.
4	Apply Scrum practices to update a Sprint Backlog during an active sprint.
4	Illustrate the difference between the Sprint Backlog and the outcomes of a Sprint Review.
4	[Scenario-Based] During a Sprint Review, stakeholders reject a delivered feature. Apply Scrum principles to determine the next course of action.
4	Explain the role of the Product Owner during Sprint Review.
4	Apply the concept of increment to explain how Sprint Review evaluates deliverables.
4	Explain the steps involved in conducting a Sprint Retrospective.
4	Apply retrospective techniques to identify improvement areas for a Scrum team.
4	Illustrate the difference between Sprint Review and Sprint Retrospective.
4	[Scenario-Based] A Scrum team consistently misses sprint goals. Apply retrospective analysis to identify the root cause and propose improvements.
4	Explain the importance of continuous improvement through retrospectives.
4	Apply the 'Start, Stop, Continue' retrospective technique to a sample Scrum team scenario.
4	Explain the responsibilities and accountability of the Scrum Master.
4	Apply the concept of self-organizing teams to describe the working of a Development Team.
4	Illustrate the interactions between the Product Owner, Scrum Master and Development Team.
4	[Scenario-Based] A Development Team faces conflicts due to unclear requirements from the Product Owner. Apply Scrum role responsibilities to resolve the issue.
4	Explain how the Scrum Master facilitates Scrum events.
4	Apply the concept of servant leadership to describe the Scrum Master's role.
4	Explain the best practices to be followed for effective Sprint Planning.
4	Apply the concept of Definition of Done to ensure quality deliverables in a sprint.
4	Illustrate how a Scrum board helps visualize project progress.
4	[Scenario-Based] A Scrum team's Daily Scrum meetings frequently exceed the time-box and lose focus. Apply Scrum best practices to improve meeting efficiency.
4	[Scenario-Based] Explain the best practices for maintaining an effective Product Backlog.
4	Apply Scrum best practices to improve collaboration within a distributed Scrum team.
5	Analyze the key components that constitute a DevOps pipeline.
5	Evaluate the benefits of adopting DevOps culture in an organization.
5	Analyze how DevOps bridges the gap between development and operations teams.
5	[Scenario-Based] An organization suffers from frequent deployment failures due to poor coordination between dev and ops teams. Analyze how DevOps adoption could resolve this.
5	[Scenario-Based] Evaluate the cultural and organizational challenges faced while implementing DevOps.
5	Analyze the role of automation in achieving DevOps goals.
5	Analyze the architecture of Docker and its core components.
5	Evaluate the advantages of containerization over traditional virtualization.
5	Analyze the process of building and running a Docker container.
5	[Scenario-Based] A development team faces 'it works on my machine' issues across environments. Analyze how Docker containerization solves this problem.
5	Evaluate the security considerations involved in using Docker containers.
5	Analyze the role of a Dockerfile in creating reproducible container images.
5	Analyze the steps involved in setting up a Continuous Integration pipeline.
5	Evaluate the impact of automated builds on software quality and delivery speed.
5	Analyze the role of source code management tools in supporting CI practices.
5	[Scenario-Based] A team experiences frequent integration conflicts due to infrequent code merges. Analyze how CI practices could reduce these conflicts.
5	Evaluate the challenges involved in adopting Continuous Integration in a legacy project.
5	Analyze the relationship between branching strategies and continuous integration.
5	Analyze the steps followed in the Test Driven Development cycle.
5	Evaluate the benefits and limitations of automated testing in a DevOps pipeline.
5	Analyze how unit testing contributes to early defect detection.
5	[Scenario-Based] A project team wants to reduce production defects by improving its testing approach. Analyze how TDD could be adopted to achieve this.
5	Evaluate the effectiveness of automated regression testing in continuous delivery.
5	Analyze the challenges of implementing TDD in a large legacy codebase.
5	Analyze the role of configuration management in maintaining consistent environments.
5	Evaluate the benefits of adopting Infrastructure as Code in DevOps practices.
5	Analyze the working of configuration management tools such as Ansible, Puppet or Chef.
5	[Scenario-Based] An organization struggles with inconsistent server configurations across environments. Analyze how configuration management tools could resolve this issue.
5	Evaluate the risks of poor configuration management in large-scale deployments.
5	Analyze the difference between manual configuration management and automated configuration management.
5	Analyze the steps involved in implementing a Continuous Deployment pipeline.
5	Evaluate the risks and benefits of fully automated deployment to production.
5	Analyze the role of automated monitoring tools in maintaining system reliability.
5	[Scenario-Based] An application experiences frequent downtime after deployments. Analyze how automated monitoring and rollback strategies could mitigate this.
5	Evaluate the importance of real-time alerting in a DevOps monitoring strategy.
5	Analyze the relationship between continuous deployment and automated monitoring in maintaining software reliability.